

Domain Modeling: Classes, Attributes, and Associations

Conceptual Classes

Conceptual classes are abstractions representing noteworthy entities in a domain. They are identified using:

- Noun phrases in **use cases**, **requirements documents**, or **domain expert knowledge**.
- Conceptual Class Category Lists.
- Domain vocabulary and real-world significance.

Example: POS Domain (iteration-1)

- Sale, CashPayment, SalesLineItem, Item, Register
- Ledger, Cashier, Customer, Store
- ProductDescription, ProductCatalog

Tip: If a real-world entity is more than a number or text, it is usually a conceptual class (e.g., Store, Airport).

Description Classes

Description classes record information about other entities:

- Separate descriptive information from individual instances.
- Avoid loss of information when instances are deleted.

Examples:

- ProductDescription describes an Item (price, text, image).
- FlightDescription describes a Flight and its route.
- ServicePlanDescription for mobile or airline packages.

UML notation: Show as a class with association to the entity it describes. Multiplicity often '1 *' (one description → many items/services).

Associations

Association: a relationship between classes that indicates a meaningful connection.

- Show only relationships we need to **remember** for some duration.
- Avoid visual noise; focus on “need-to-remember” associations.

- Use ClassName-VerbPhrase-ClassName format (e.g., **Sale Paid-by Cash-Payment**).
- Multiplicity shows number of associated instances ('1', '0..1', '*', '1..*').

Common Associations (examples in POS domain):

- Sale Paid-by CashPayment
- Sale Contains SalesLineItem
- SalesLineItem Records-sale-of Item
- Register Contains Item
- Cashier Uses Register

UML Hint: Each association has optional roles, multiplicity, and optional reading direction arrow.

Attributes

Attributes: logical data values of a class, satisfying information needs in scenarios.

- Include attributes implied by requirements or use cases.
- Prefer primitive or simple data types (Boolean, Number, Date, Time, String, Character).
- Use data type classes for quantities, units, or structured identifiers (e.g., Money, ItemID, Address).
- Derived attributes: prefix '/' (e.g., '/total' in Sale).

POS Domain Example:

- Sale: dateTime
- Store: name, address
- Cashier: ID
- SalesLineItem: quantity
- Payment: amount : Money
- ProductDescription: itemID : ItemID, price, description
- CashPayment: amountTendered

Tip: Do not use foreign keys as attributes; relate classes using associations instead.